Default Labels, Switching Costs, and Mixed Train-Free Mitigation in LLM Pair Decisions

Anonymous submission

Abstract

Default and current-option labels can change advice even when the underlying alternatives are unchanged, yet deployed decision interfaces often combine those labels with legitimate switching costs, making it unclear whether selecting the current option is a bias or a rational cost-sensitive choice. We study this interaction with a train-free pair-decision benchmark built from MovieLens, Yelp Review Full, and Goodbooks-10K items, first evaluated with Qwen/Qwen2.5-7B-Instruct under seven prompting or aggregation conditions and then stress-tested with a usable Llama 3.1 8B replication. This separation makes the result diagnostic rather than a universal method win: direct prompting shows family-dependent labeled-option pull, and label-blind cost-aware aggregation preserves switching-cost override rows, but the same cost reinsertion fails badly on Yelp non-overrides and does not transfer as a robust mitigation across backends. In the Qwen 7B primary run, the cost-aware variant reaches 94.2% on Goodbooks versus 89.1% for the no-cost ablation, but reaches only 49.0% on Yelp versus 68.8% for no-cost and 63.4% for direct prompting. In the Llama replication, prompted debiasing falls below raw direct prompting in all three families, while label-blind no-cost remains at least 93.2%. The evidence supports a calibrated, row-level, slice-aware benchmark and analysis: train-free cost reinsertion can protect legitimate overrides, but it needs a confidence gate and broader validation before it can be described as a reliable mitigation.

Introduction

Many decision interfaces show a current, default, or recommended option before a user asks for advice. A travel site may mark one itinerary as the current reservation, a recommender may display an item as already selected, and a support tool may identify a default plan. Human decision research shows that defaults can affect choices (Samuelson and Zeckhauser 1988), that default framing can alter consequential enrollment decisions (Johnson and Goldstein 2003), and that reference-dependent preferences interact with loss aversion and switching costs (Kahneman, Knetsch, and Thaler 1991; Tversky and Kahneman 1991). These effects create a practical ambiguity for language-model advice: selecting the current option may be a presentation bias, but it may also be correct if switching is genuinely costly.

Recent LLM work gives reasons to expect this ambiguity to matter. Models can be sensitive to answer-position and selection effects (Zheng et al. 2023), recommendation-position effects (Bito, Ren, and He 2025), order effects (Yin, Vardi, and Choudhary 2025), and sycophantic user cues (Sharma et al. 2023). Prompting and inference-time interventions can reduce some surface-form effects, but they do not by themselves distinguish an illegitimate default label from a legitimate cost-adjusted preference. A method that simply removes labels may undercount valid switching costs, while a method that always restores costs may over-preserve the displayed current option.

This paper studies that distinction in a controlled pair-decision setting. Each example has two alternatives, a displayed label condition, a no-cost merit target, and a cost-adjusted target. We evaluate seven train-free conditions: direct prompting, prompted debiasing, order-shuffle voting, a RISE-style position-robust adaptation, label-blind aggregation with cost reinsertion, label-blind aggregation without cost reinsertion, and a label-only control. The central comparison is between the two label-blind variants, because they share the same utility-estimation step and differ only in whether switching cost is reintroduced.

The answer is mixed in a way that matters scientifically. In the Qwen 7B primary run, direct prompting has labeled-choice rates of 55.4% on MovieLens, 45.7% on Yelp Review Full, and 53.6% on Goodbooks-10K. Cost-aware label-blind aggregation improves over direct prompting on MovieLens and Goodbooks by paired sign tests, and it preserves override rows in all three families. It is not uniformly better, however. On Yelp, cost-aware aggregation scores 49.0%, below no-cost aggregation at 68.8% and direct prompting at 63.4%. A Llama 3.1 8B replication sharpens the warning: prompted debiasing is below raw direct prompting on all three families, and label-blind no-cost reaches 93.3%, 93.3%, and 93.2% on MovieLens, Yelp, and Goodbooks. The strongest negative result is therefore part of the contribution, not a caveat hidden after an average.

The paper makes four contributions. First, it defines a benchmark protocol that separates displayed default labels, no-cost merit labels, and cost-adjusted labels. Second, it evaluates seven train-free decision conditions under a primary Qwen 7B backend and adds a Llama 3.1 8B replication check, reporting aggregate, paired-test, slice, and backend-delta evidence rather than a single pooled leaderboard. Third, it shows that cost reinsertion is beneficial

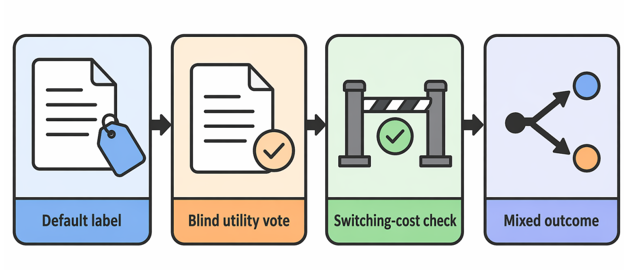

Figure 1: Default-label diagnostic: the benchmark separates option merit, default/current labels, and switching-cost checks so the analysis can distinguish labeled-option bias from legitimate cost preservation.

for legitimate override rows but harmful on important Yelp non-overrides, and that prompted-debiasing robustness is rejected by the Llama replication. Fourth, it identifies the missing evidence needed for a stronger mitigation claim: confidence-gated cost reinsertion, deeper Yelp failure taxonomy, user-versus-system label-source ablations, and additional valid instruction-tuned backend checks.

Related Work

The paper is grounded in status quo bias and reference-dependent choice. Defaults can become sticky even without explicit evidence that the default is best (Samuelson and Zeckhauser 1988). Related evidence shows that default framing can alter consequential decisions (Johnson and Goldstein 2003). Loss aversion provides one mechanism by which switching costs enter decisions (Kahneman, Knetsch, and Thaler 1991), and reference-dependent models distinguish utility changes from presentation changes (Tversky and Kahneman 1991). Our benchmark operationalizes this distinction by keeping merit labels and cost-adjusted labels separate.

LLM bias work has studied cognitive bias in recommendation and decision settings (Lyu et al. 2024; Pilli and Nallur 2026). Position and ordering work shows that recommendation order and judge input order can change outputs (Bito, Ren, and He 2025; Xu et al. 2026; Petrov, Murtagh, and Nagesh 2025). Prompt-based reranking and talent-recommendation studies motivate train-free or inference-time interventions but do not isolate switching-cost reinsertion (Islam and Faruk 2025; Du and Liu 2026). Other work studies fragile preferences (Yin, Vardi, and Choudhary 2025), sycophancy (Wang et al. 2025; Sharma et al. 2023), cognitive-bias mitigation (Huang et al. 2026), and activation-steering debiasing (Li et al. 2025). Multiple-choice robustness work shows that surface form can alter model choices (Zheng et al. 2023). Our setting differs by pairing the label with a cost model and by testing a no-cost versus cost-aware aggregation ablation, which exposes the tradeoff between debiasing and valid cost retention.

Order sensitivity is broader than recommendation prompts. Prior work reports position effects in summarization, context use, premise order, prompt order, and multi-constraint instruction following (Chhabra, Askari, and Mohapatra 2024; Ravaut et al. 2023; Chen et al. 2024; Guan et al. 2025; Zeng et al. 2025). Other studies propose direct position-bias mitigation, self-supervised debiasing, or stronger symbol binding for multiple-choice selectors (Yu et al. 2024; Liu et al. 2024; Xue et al. 2024; Wang et al. 2024). These papers motivate the order-shuffle and RISE-style controls in our matrix, but they do not answer whether a displayed current option should be preserved when a switching-cost label is legitimate.

Inference-time debiasing work also motivates a conservative claim scope. BiasFilter, teacher-student permutation debiasing, permutation-aware reinforcement learning, and reward-model interventions all change inference-time behavior without retraining the evaluated LLM from scratch (Cheng et al. 2025; Liusie, Fathullah, and Gales 2024; Zheng et al. 2026; Shinoda, Nishida, and Nishida 2026). Mechanistic analyses of bias and in-context label behavior provide a complementary route for locating biased computations (Golgoon, Filom, and Kannan 2024; Chandna, Bashir, and Sen 2025; Yu and Ananiadou 2024; Li and Gao 2024). Our contribution is not a new white-box intervention; it is a row-level benchmark that shows when a cheap train-free rule preserves valid cost overrides and when it damages no-cost utility choices.

Two design choices separate this study from broader bias surveys. First, the benchmark makes the displayed label experimentally controllable: it is not inferred from historical user behavior or ranking position, unlike many recommendation-bias settings (Bito, Ren, and He 2025; Islam and Faruk 2025). Second, the benchmark keeps both a merit target and a cost-adjusted target, following the distinction between reference effects and utility changes in classic decision work (Kahneman, Knetsch, and Thaler 1991; Tversky and Kahneman 1991). That makes it possible to identify cases where following the current option is correct because switching is costly, rather than treating every current-option choice as bias. This distinction is the reason the paper keeps both positive and negative cost-aware findings in the main results.

Recommendation and reranking papers often study position effects, presentation bias, or fairness interventions after a ranking model has already produced candidates (Lyu et al. 2024; Du and Liu 2026). Here the evaluation unit is simpler: a pair prompt with two alternatives and a controlled label/cost perturbation. The simplicity is a limitation for deployment realism, but it is useful diagnostically because a row-level score can be traced to a specific source pair, displayed label, order, cost condition, method, and raw scored response.

Benchmark and Method

Each task presents two items and asks for a choice. The benchmark constructs three source families: movie recommendations from MovieLens, review-quality judgments from Yelp Review Full, and book recommendations from Goodbooks-10K. Conditions vary whether a label names an option as current/default/recommended, whether the order is

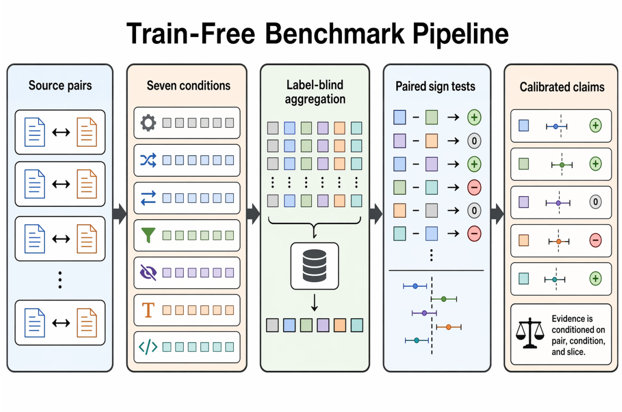

Figure 2: Pipeline overview: source pairs are transformed into label and cost conditions, scored by a primary Qwen2.5-7B backend with a Llama 3.1 8B replication check, aggregated under seven train-free conditions, and evaluated with aggregate, paired, slice, and backend-delta evidence.

AB or BA, and whether a switching cost should change the cost-adjusted gold label.

The source pairs are designed to make the decision target explicit. For every row, the no-cost merit target identifies which option should be chosen if switching is free. The cost-adjusted target then asks whether a modeled switching cost should change that choice. The displayed label is generated independently as a presentation variable: it can point to the current or default option, but it is not itself evidence that the option is better. This lets the benchmark separate three effects that collapse together in ordinary interface logs: content preference, label attraction, and cost preservation.

The evaluated conditions are direct prompting, an explicit debiasing prompt, order-shuffle voting, a RISE-style position-robust adaptation, label-blind aggregation with switching-cost reinsertion, label-blind aggregation without cost reinsertion, and a label-only condition. The core mechanism is label-blind aggregation: remove labels while estimating option utility, aggregate order-balanced choices, and optionally reinsert switching cost only when the cost-adjusted gold label requires it. This final reinsertion step is the mechanism under test; the evidence shows it should be treated as conditional.

The benchmark deliberately separates three labels that are often conflated in prose. The merit label asks which option is better if switching costs are ignored. The cost-adjusted label asks which option should be selected after the switching cost is applied. The displayed default or current label is a presentation variable, not evidence of merit. This separation is what makes the Yelp failure informative: a rule that is useful for legitimate override rows can still over-preserve labeled options when the no-cost utility vote already matches the cost-adjusted target.

The data generators also mirror order. Each source pair can appear as AB or BA, so a method that only tracks left/right position is exposed by the order-shuffle and label-blind comparisons. This is important for the paper's interpretation because a high score under direct prompting might reflect content understanding, positional artifacts, label attraction, or cost sensitivity. The seven-condition matrix is designed to separate those mechanisms enough for a diagnostic claim, while leaving broader causal claims to later ablations.

MovieLens and Goodbooks rows instantiate recommendation-style preference judgments, while Yelp rows instantiate review-quality judgments with larger textual contrasts. The Yelp family is therefore not merely another copy of the same task template: it is the family where cost-aware reinsertion most clearly damages non-override rows. This heterogeneity prevents the method section from collapsing into a single positive aggregate and forces the paper to report where the mechanism fails.

The label-blind variants are implemented without training. They use the same source rows as direct prompting but remove the presentation label before utility aggregation. The no-cost variant stops after utility aggregation, whereas the cost-aware variant reintroduces switching cost. The comparison between those two variants is the cleanest test of whether cost reinsertion itself helps or hurts under a fixed utility estimate.

The seven conditions cover three roles. Direct prompting and prompted debiasing test whether a simple instruction changes the baseline decision. Order-shuffle voting and the RISE-style adaptation test whether order-balanced aggregation is enough to reduce surface-form effects. The two label-blind variants and the label-only control isolate the displayed-label mechanism: the no-cost variant tests label removal alone, the cost-aware variant tests cost reinsertion after label removal, and the label-only condition estimates how far label structure can go without item content. This decomposition is why the paper can report a negative cost-aware result without treating it as a failed experiment.

Operationally, the two generative baselines are the original pair prompt and the same prompt prefixed with an instruction to ignore arbitrary current/default/recommended labels unless explicit switching cost makes staying rational. Order-shuffle voting reuses valid direct-prompt choices and takes a majority over mirrored order variants for the same source pair, label variant, and cost slice. The RISE-style control also reuses direct-prompt rows, but aggregates over label variants within each source pair and cost slice and is reported only as a position-robust adaptation. Label-blind aggregation estimates utility from no-label rows; the no-cost arm reports that utility vote, while the cost-aware arm restores the labeled option only for large-cost rows when the displayed label disagrees with the utility vote and the utility gap is at most 1.0. Label-only keeps item content out of the decision and measures how far label structure alone can go.

The row identifiers preserve the source family, source-pair identity, label condition, order condition, and cost condition. This matters because the paired tests are only meaningful when two methods are compared on the same underlying decision. The benchmark therefore does not compare an aggregate score from one set of prompts against a score from another unrelated set. It compares common task identifiers after each method has produced a scored choice for that identifier.

The label-only condition is deliberately weak as a decision method but useful as a diagnostic control. It asks how much of the target can be recovered from the displayed label pattern alone, without item content. If a full method is close to the label-only control, the result suggests that presentation structure is carrying too much of the answer. If a label-blind method beats the label-only control on one family but not another, the benchmark can separate content utility from label attraction rather than averaging them together.

The RISE-style and order-shuffle conditions play a different role. They test whether a surface-order intervention is enough before the paper introduces label blindness and cost reinsertion. A method that simply votes across AB and BA variants can reduce left/right artifacts, but it cannot decide whether a current-option label should be treated as illegitimate presentation information or as a cue to a real switching cost. That is why order controls appear in the same matrix as label-blind variants rather than replacing them.

Experimental Setup

The evaluated system is a fixed pair-decision harness, not a learned model or a deployed recommender. For each source pair, the harness creates label, order, and switching-cost variants; sends the generative conditions to the primary Qwen/Qwen2.5-7B-Instruct backend and a Llama 3.1 8B replication backend; scores the selected option against the cost-adjusted target; and applies deterministic aggregation controllers for order-shuffle voting, RISE-style label aggregation, label-blind no-cost voting, label-blind cost reinsertion, and the label-only diagnostic. Thus the seven rows in Table 1 are method roles within one controlled benchmark: two prompting baselines, two order/label aggregation controls, two label-blind ablations that isolate cost reinsertion, and one content-free label diagnostic. The reported metrics are cost-adjusted accuracy, labeled-choice rate for direct prompting, paired sign tests on common task identifiers, override/non-override slice accuracy, and backend deltas for the usable replication.

The primary Qwen 7B evaluation contributes 120,960 method rows across three families and seven methods. A Llama 3.1 8B replication contributes another 120,960 rows and is used for backend-delta claims. A Qwen/Qwen2.5-1.5B base-model run also completed 120,960 rows, but all 43,200 generative rows were invalid; the paper treats that run only as negative backend-suitability evidence, not as robustness support. Generative methods use deterministic decoding with temperature 0, a maximum of 2 generated tokens, newline stopping, and a 768-token context limit; aggregate-only methods reuse the same scored direct-prompt rows and do not make additional model calls. The paper-facing metric is cost-adjusted accuracy, with labeled-choice rate used as the diagnostic for default-label pull. Pairwise comparisons use exact paired sign tests over common task identifiers. The paper does not claim method-level multi-model robustness because the usable replication contradicts several positive Qwen-only interpretations.

The evidence is organized at three levels. Aggregate rows report family-by-method cost-adjusted accuracy and labeled-choice rate. Paired tests compare methods on common task identifiers rather than treating rows as independent across methods. Slice tables then split label-blind cost and no-cost variants into override, non-override near-tie, and wide-gap regimes. The interpretation follows this hierarchy: default-label bias and override preservation are supported, universal cost-aware superiority is rejected, and broad robustness remains untested.

Invalid rows are tracked in the aggregate schema rather than silently dropped from the evidence story. The aggregate table records valid and invalid counts for every family-method row. The analysis focuses on accuracy and labeled-choice behavior because the key contradiction is not an invalid-output artifact; it appears in valid scored comparisons between cost-aware and no-cost aggregation.

Results

Table 1 rejects a universal cost-aware superiority claim. Cost-aware aggregation beats direct prompting on MovieLens and Goodbooks by paired sign tests: MovieLens has 969 wins versus 490 losses ($p = 1.4450177393288226e-36$), and Goodbooks has 926 wins versus 169 losses ($p = 7.0376379721151294e-127$). Yelp reverses the story: the stored direct-versus-cost comparison has 778 wins and 1816 losses against cost-aware aggregation.

The no-cost ablation is the sharpest diagnostic. On MovieLens, cost-aware and no-cost are statistically close at the aggregate sign-test level ($p = 0.12307327589626703$). On Yelp, no-cost strongly outperforms cost-aware with 1906 rows favoring no-cost in the stored comparison. On Goodbooks, cost-aware is clearly better than no-cost. The result is therefore family-dependent, not a global mitigation law.

Prompted debiasing is also informative but not decisive as a mitigation story. It is strong on MovieLens and Goodbooks and modestly improves Yelp over direct prompting, yet it does not isolate whether the model changed because it ignored the label, followed an instruction to be fair, or shifted its response style. The label-blind variants are more diagnostic because the cost-aware and no-cost arms differ only in whether the switching-cost pass is reinserted after utility aggregation.

The rejected universal claim matters for paper framing. A deployment-facing mitigation paper would need the cost-aware pass to dominate no-cost and direct alternatives across families or to specify a reliable gate for when reinsertion is appropriate. The current evidence instead supports a benchmark-and-diagnostic paper: it identifies where default labels pull choices, where cost preservation is legitimate, and where train-free cost reinsertion can be actively harmful.

The family-level pattern is also important. MovieLens is the setting where a simple instruction is most effective: prompted debiasing reaches 92.7%, above both label-blind variants. This suggests that, for short recommendation-style pairs, a direct instruction can reduce the displayed-label effect without needing a separate cost-reinsertion mechanism. The label-blind cost-aware condition is still above direct prompting, but it does not beat the prompted baseline or the no-cost label-blind variant.

Table 1: Main setup/result matrix: all 21 family–method cells expose source, task count, method role, backend or reuse policy, metric, and key scores.

Family/source	Method role	Rows	Backend/budget	Metric	Acc.	Label rate
MovieLens pairs	direct baseline	7,200	Qwen2.5; temp 0; 2 tok.	cost acc.	81.1%	55.4%
MovieLens pairs	prompted debias	7,200	Qwen2.5; temp 0; 2 tok.	cost acc.	**92.7%**	50.3%
MovieLens pairs	order-shuffle vote	3,600	scored-row reuse; order vote	cost acc.	63.9%	–
MovieLens pairs	RISE-style control	720	scored-row reuse; RISE-style	cost acc.	91.8%	–
MovieLens pairs	label-blind + cost	7,200	scored-row reuse; cost rule	cost acc.	87.8%	–
MovieLens pairs	label-blind no-cost	7,200	scored-row reuse; no-cost	cost acc.	88.5%	–
MovieLens pairs	label-only control	7,200	label structure only	cost acc.	86.4%	–
Yelp Review Full pairs	direct baseline	7,200	Qwen2.5; temp 0; 2 tok.	cost acc.	63.4%	45.7%
Yelp Review Full pairs	prompted debias	7,200	Qwen2.5; temp 0; 2 tok.	cost acc.	65.1%	49.5%
Yelp Review Full pairs	order-shuffle vote	3,600	scored-row reuse; order vote	cost acc.	37.6%	–
Yelp Review Full pairs	RISE-style control	720	scored-row reuse; RISE-style	cost acc.	63.1%	–
Yelp Review Full pairs	label-blind + cost	7,200	scored-row reuse; cost rule	cost acc.	49.0%	–
Yelp Review Full pairs	label-blind no-cost	7,200	scored-row reuse; no-cost	cost acc.	**68.8%**	–
Yelp Review Full pairs	label-only control	7,200	label structure only	cost acc.	60.3%	–
Goodbooks-10K pairs	direct baseline	7,200	Qwen2.5; temp 0; 2 tok.	cost acc.	83.7%	53.6%
Goodbooks-10K pairs	prompted debias	7,200	Qwen2.5; temp 0; 2 tok.	cost acc.	89.6%	49.1%
Goodbooks-10K pairs	order-shuffle vote	3,600	scored-row reuse; order vote	cost acc.	69.4%	–
Goodbooks-10K pairs	RISE-style control	720	scored-row reuse; RISE-style	cost acc.	91.4%	–
Goodbooks-10K pairs	label-blind + cost	7,200	scored-row reuse; cost rule	cost acc.	**94.2%**	–
Goodbooks-10K pairs	label-blind no-cost	7,200	scored-row reuse; no-cost	cost acc.	89.1%	–
Goodbooks-10K pairs	label-only control	7,200	label structure only	cost acc.	89.5%	–

Yelp is different. Direct prompting and prompted debiasing are close, at 63.4% and 65.1%, while order-shuffle voting drops to 37.6%. The cost-aware label-blind variant is lower than direct prompting, prompted debiasing, label-only, and the no-cost label-blind ablation. This means the negative result is not a small perturbation of one baseline. It is a broad warning that cost reinsertion can override useful no-cost utility evidence in text-heavy review-quality comparisons.

Goodbooks provides the strongest positive case for cost reinsertion. Cost-aware aggregation reaches 94.2%, above prompted debiasing at 89.6%, no-cost aggregation at 89.1%, and label-only at 89.5%. The result shows that the method can help when the switching-cost target aligns with the utility estimate often enough to preserve override rows without imposing a large non-override penalty. Taken with Yelp, the correct conclusion is not that cost reinsertion is weak; it is that it is a conditional operation whose failure mode must be measured.

The order and RISE-style controls help interpret this conclusion. Order-shuffle voting is weak in all three families, especially Yelp, indicating that a simple majority vote over order variants is not enough to recover the cost-adjusted target. The RISE-style adaptation is stronger on MovieLens and Goodbooks but remains below the best family-specific method. These controls make the label-blind comparison more informative: the cost/no-cost difference cannot be explained only by left/right order effects.

Backend replication. The Llama 3.1 8B replication changes the scientific reading from a single-backend mitigation story to a backend-dependent diagnostic. Prompted debiasing falls below raw direct prompting on every family: 68.2% versus 78.9% on MovieLens, 49.8% versus 50.2% on Yelp, and 55.9% versus 63.1% on Goodbooks. Label-blind cost-aware aggregation also weakens sharply relative to the Qwen 7B primary run, while label-blind no-cost is consistently high at 93.3%, 93.3%, and 93.2% on MovieLens, Yelp, and Goodbooks. These backend deltas reject a robustness claim for prompted debiasing or automatic cost reinsertion, but they support keeping no-cost label-blind aggregation as a strong baseline that every future cost-aware method must beat.

The Qwen 1.5B base-model run is not counted as positive replication. It completed the benchmark harness, but all 43,200 generative rows for raw direct and prompted debiasing were invalid, so it is evidence about backend suitability and extraction failure rather than about method effectiveness. The paper therefore uses two backends for substantive multi-backend interpretation: Qwen 7B and Llama 3.1 8B.

Table 3 also shows why statistical testing does not rescue the universal claim. Several comparisons are overwhelmingly different, but their directions disagree across families. The paired tests therefore strengthen the mixed conclusion: the evidence is not noisy around a single positive story; it is decisive in different directions depending on the family and slice.

The paired-test directions also determine what the paper can claim. MovieLens and Goodbooks support a cost-aware improvement over direct prompting, with many more wins than losses under common task identifiers. Yelp has the opposite direction: the direct-versus-cost comparison records more losses than wins for cost-aware aggregation. The no-cost versus cost-aware comparison then isolates the reinsertion step. The near-null MovieLens result, negative Yelp result, and positive Goodbooks result together rule out a single global statement about the method.

The family split is not a nuisance variable. MovieLens and Goodbooks instantiate recommendation-style decisions in which small preference differences and switching costs can plausibly interact. Yelp uses review-quality text, so a wide semantic gap between the alternatives can make cost reinsertion look especially brittle. The same train-free mechanism is therefore tested under source families that differ in how easy it should be to preserve a no-cost utility judgment. This is why the paper reports family-level tables instead of pooling all rows into one headline accuracy.

The main table also shows that no single cheap control explains the results. Prompted debiasing is strong on MovieLens, while label-blind cost-aware aggregation is strongest on Goodbooks. Yelp favors no-cost aggregation, not either order-shuffle voting or cost-aware reinsertion. If the only problem were answer order, the order-based methods would be expected to dominate the relevant families. If the only problem were missing switching costs, the cost-aware variant would not collapse on Yelp. The measured pattern instead supports a decomposition: default labels, order, content utility, and switching cost are separate variables that need to be audited separately.

This decomposition changes how the negative result should be read. A failed mitigation story would treat Yelp as an exception to be smoothed away. A diagnostic benchmark treats Yelp as the stress case that reveals the missing gate. The cost-aware rule can be correct on override rows, but it should not be allowed to override stable no-cost utility evidence in non-override rows. The current benchmark cannot yet learn that gate, but it identifies the exact comparison a future gate must satisfy.

Slice Analysis and Failure Cases

The slice table explains why the mixed aggregate occurs. Cost-aware aggregation preserves override rows in every family, with deltas of 91.7, 60.8, and 80.8 points over no-cost on MovieLens, Yelp, and Goodbooks. But non-override rows are not free: Yelp loses 29.1 points on near ties and 22.5 points on wide-gap rows.

The sampled Yelp losses show large-cost/default-label cases where the cost-aware rule keeps or restores the labeled option even when the no-cost utility vote matches the cost-adjusted gold label. The current taxonomy is sampled, not exhaustive, so the paper treats the failure mechanism as partially characterized rather than fully explained.

The slice structure explains why aggregate accuracy alone is misleading. Override rows are the rows where switching cost should change the no-cost decision, and cost-aware aggregation reaches 100.0% on those rows in all three families. Non-overrides expose the cost of that mechanism: Yelp loses 29.1 points on near ties and 22.5 points on wide-gap rows, so the penalty is not confined to ambiguous choices. The sampled Yelp losses make this concrete without replacing the aggregate evidence: the supplemental rows include clear review-quality contrasts where cost-aware aggregation is wrong and no-cost aggregation is right. The paper therefore treats the failure mechanism as partially characterized and the contribution as a measurement of the tradeoff, not as a solved mitigation.

Discussion

The main methodological lesson is that label blindness and cost awareness are not the same intervention. Label blindness asks whether the model can estimate utility without being pulled by a displayed marker; cost awareness asks whether a legitimate switching cost should override that estimate. The Yelp and Llama results show that combining the two requires a gate: if the label-blind utility estimate is already decisive, automatic cost reinsertion can preserve the wrong option.

This is why the paper keeps aggregate, paired, slice, and backend-delta evidence together. A pooled leaderboard would hide the rows where a proposed method fails, while a slice-only story would not show whether the aggregate comparison is stable. The fixed reading order is therefore part of the benchmark design: first compare family-method rows, then check paired directions, then inspect override and non-override slices, and finally ask whether the pattern transfers to another valid instruction-tuned backend.

The practical recommendation is procedural rather than deployable. Before accepting a default-label mitigation claim, report a no-cost ablation, a cost-aware variant, order-sensitive controls, backend deltas, and override/non-override slices. A future positive method should preserve override-row gains while reducing Yelp-style non-override damage; a method that improves only the average or only one family has not solved the problem exposed here.

Backend-transfer audit. Table 2 makes the backend evidence visible rather than leaving it as prose. The largest

Table 2: Backend-transfer audit: Llama weakens prompted debiasing by 15.3–33.7 points, while no-cost label-blind aggregation remains at least 93.2%.

Family	Prompt Δ	Cost Δ	Llama no-cost	1.5B invalid
MovieLens	-24.5 pp	-13.6 pp	93.3%	14.4k/14.4k
Yelp	-15.3 pp	-35.1 pp	93.3%	14.4k/14.4k
Goodbooks	-33.7 pp	-47.6 pp	93.3%	14.4k/14.4k

Table 3: Paired sign tests: cost-aware aggregation is supported on MovieLens and Goodbooks against direct prompting, but Yelp contradicts a universal win and strongly favors the no-cost ablation.

Family	Test	Wins	Losses	Ties	p
MovieLens	Direct–prompt	1,020	187	5,993	$3.6e{-139}$
MovieLens	Direct–cost	969	490	5,741	$1.4e{-36}$
MovieLens	No-cost–cost	480	530	6,190	0.123
Yelp	Direct–prompt	1,070	949	5,181	0.0076
Yelp	Direct–cost	778	1,816	4,606	$1.0e{-94}$
Yelp	No-cost–cost	480	1,906	4,814	$4.9e{-200}$
Goodbooks	Direct–prompt	711	289	6,200	$1.2e{-41}$
Goodbooks	Direct–cost	926	169	6,105	$7.0e{-127}$
Goodbooks	No-cost–cost	480	108	6,612	$6.6e{-57}$

negative transfer is not a small formatting or parser artifact: Goodbooks prompted debiasing loses 33.7 points under Llama, and Goodbooks cost-aware aggregation loses 47.6 points relative to the Qwen 7B primary run. Yelp shows the same direction for cost reinsertion, with a 35.1-point loss. These numbers are why the paper rejects a method-general robustness claim even though the Qwen 7B primary run contains strong positive slices.

The table also separates two different failure modes. Llama is a valid instruction-tuned replication backend, so its no-cost label-blind scores of 93.3%, 93.3%, and 93.3% are substantive evidence that the no-cost baseline is strong. Qwen 1.5B is different: every generative row for raw direct and prompted debiasing is invalid in each family. Counting that run as robustness support would be misleading. Its role is only to show that the harness detects a backend that cannot reliably emit parseable pair choices under the fixed answer format.

This audit scopes away the planned ablations instead of pretending they are complete. A confidence-gated cost-reinsertion method would need to decide when the no-cost utility vote is decisive enough to block cost reinsertion. A user-versus-system label-source split would need to test whether the displayed current option came from the user, the interface, or a recommendation model. A broader failure taxonomy would need to sample more than the current Yelp loss set. The current paper can therefore support a diagnostic benchmark and a negative transfer result, but it cannot claim that the missing gate has already been learned.

Conclusion

Default/current/recommended labels should not be evaluated only by asking whether a model follows the label, because following the current option can be either an error or the correct cost-adjusted decision. This paper separates displayed labels, no-cost utility labels, and cost-adjusted labels so those cases can be measured rather than averaged together. The resulting evidence is deliberately mixed: train-free cost-aware aggregation protects legitimate override rows in the Qwen 7B primary run, but it fails as a universal mitigation because Yelp non-overrides suffer and the Llama replication favors no-cost label-blind aggregation. The benchmark therefore supplies a diagnostic frame for future variants: preserve override protection, reduce Yelp-style non-override damage, keep the no-cost ablation visible, and repeat the test across valid instruction-tuned backends before claiming general mitigation.

Limitations

The benchmark uses controlled pair decisions rather than real interactive decision support, so it does not model preference uncertainty, repeated interaction, personalization, or learned switching costs. The usable replication covers one additional backend and contradicts several Qwen-only positive interpretations, so the paper cannot claim multi-model robustness. The current analysis identifies sampled Yelp failures but does not provide a complete taxonomy, and it does not test a confidence-gated cost-reinsertion variant. These limits are the reason the paper frames the contribution as a diagnostic benchmark and evidence analysis, not as a deployable decision policy.

The source families also cover a narrow form of advice. MovieLens and Goodbooks are recommendation-like, while Yelp uses review-quality judgments; none of the families represents medical, financial, legal, or safety-critical advice. The label conditions are constructed so that merit labels and cost-adjusted labels are observable, which is useful for measurement but simpler than real interfaces where defaults may be personalized, strategic, stale, or user-provided. The benchmark should therefore be used to audit mechanisms

Figure 3: Aggregate and slice result: the left panel shows that Goodbooks supports cost-aware aggregation while Yelp contradicts a universal superiority claim; the right panel shows that override protection coexists with Yelp non-override penalties.

Table 4: Slice result: override preservation is real, but Yelp non-overrides carry the largest cost-aware penalty.

Family	Slice	Rows	Cost	No-cost	Delta
MovieLens	Override	480	100.0%	8.3%	+91.7 pp
MovieLens	Near-tie non-override	3,120	72.1%	87.8%	-15.7 pp
MovieLens	Wide-gap	3,600	99.7%	99.7%	+0.0 pp
Yelp	Override	480	100.0%	39.2%	+60.8 pp
Yelp	Near-tie non-override	3,120	48.1%	77.2%	-29.1 pp
Yelp	Wide-gap	3,600	43.1%	65.6%	-22.5 pp
Goodbooks	Override	480	100.0%	19.2%	+80.8 pp
Goodbooks	Near-tie non-override	3,120	88.0%	87.9%	+0.1 pp
Goodbooks	Wide-gap	3,600	98.9%	99.4%	-0.6 pp

before broader deployment studies, not to infer how all default labels behave in the wild.

The evidence gaps are specific. A stronger claim requires additional valid instruction-tuned backend replications beyond the mixed Qwen/Llama evidence, a user-versus-system label-source split, a confidence-gated cost-aware variant, a broader Yelp failure taxonomy, and uncertainty estimates beyond the exact paired tests reported here. Until those checks are run, the paper cannot claim that cost reinsertion is reliable across models, domains, or interface sources.

The current evidence also rules out three tempting shortcuts. First, it is not enough to report the best family, because the same cost-aware rule is 94.2% on Goodbooks and 49.0% on Yelp. Second, it is not enough to report an instruction-only debiasing prompt, because Llama reverses that story by putting prompted debiasing below raw direct prompting in all three families. Third, it is not enough to add a smaller backend as a nominal replication, because the Qwen 1.5B probe completed rows but produced invalid generative outputs for every raw-direct and prompted-debias row. These are not formatting caveats; they are measurement constraints on the claim. They motivate the paper's narrowed wording and define the minimum repair path for a stronger future result.

Responsible Use

The evaluated rule should not be used to make user-facing recommendations without additional validation. A default-aware assistant could over-preserve a displayed option when cost information is treated as too strong a prior, especially in settings with consequential user costs. The safer use of this benchmark is as an audit: report direct prompting, no-cost and cost-aware label-blind variants, order-sensitive controls, paired tests, and override/non-override slices before describing a mitigation as reliable.

The main responsible-use risk is overinterpreting a current/default/recommended label as normative evidence. In a real decision-support setting, that could preserve an option because it is already selected rather than because it is better for the user under a stated cost model. Conversely, removing all cost information can be harmful when switching is genuinely costly. Any deployed system would need explicit provenance for the label source, a calibrated estimate of switching cost, and a policy for abstaining or asking for clarification when those signals disagree.

The reporting obligation follows directly from the measured failure modes. A paper or product note that reports only the Qwen 7B Goodbooks gain would omit the Yelp non-override loss, the Llama negative transfer, and the invalid Qwen 1.5B probe. Conversely, a note that reports only the Llama no-cost strength would omit the legitimate override rows where cost-aware aggregation reaches 100.0% in all three families. The responsible claim is therefore a complete audit matrix: family-level aggregate scores, paired direction tests, override and non-override slices, valid-backend deltas, and invalid-output counts. This is also the minimum evidence package future confidence-gated or label-source variants should beat before being described as reliable.

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Reproducibility Checklist

The reproducibility materials record the benchmark families, method rows, model/backends, aggregate tables, paired sign tests, figure source data, multi-run backend deltas, and raw scored rows. All numeric claims in the manuscript are mapped to generated aggregate, significance, bias, slice, multi-run, or backend-delta evidence. The Qwen 1.5B base-model probe is recorded as invalid-output backend-suitability evidence rather than robustness support.

Appendix: Evidence Gaps and Supplemental Tables

The supported claims are default-label bias, Qwen 7B override preservation, and Llama no-cost label-blind strength. Universal cost-aware superiority, prompted-debiasing backend robustness, and cost-aware aggregation being free on non-overrides are rejected. Sufficiency for a strong long-paper method claim remains weak. Missing evidence includes additional valid instruction-tuned backend replication, user-versus-system label-source ablation, confidence-gated cost-aware reinsertion, broader Yelp failure taxonomy, and uncertainty estimates for generality.